

Math 113 – Lecture 03

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1 Subgroups (Section 5)

1.1 Basic Relationships between Subgroups

Definition 1.1 (Subgroup). A subgroup H of a group G is a subset, closed under the operation $*$ in G , such that $\langle H, * \rangle$ is a group. Write

$$H \leq G \quad (H \subseteq G).$$

Example 1.2 (Subgroup Examples). Take $H = (\mathbb{Z}, +)$ and $G = (\mathbb{Q}, +)$.

Example 1.3 (NOT A GROUP). Taking:

$$H = (\mathbb{Z}, \cdot) \quad \text{and} \quad G = (\mathbb{Q}, +)$$

Must use induced operation $(+)$ from G .

Example 1.4 (Smallest Subgroup of G). Take e .

It's closed as $e * e = e$, in fact it is the trivial group.

Example 1.5 (G is a subgroup of itself). $G \subseteq G$.

Theorem 1.6 (Groups inside other groups) If $K \leq H$ and $H \leq G$, then $K \leq G$.

Proof. Since $K \subseteq H \subseteq G$, we know that $K \subseteq G$.

Since $K \leq H$, we know that K is closed under $*$, and $\langle K, * \rangle$ is a group, so $K \leq G$. □

1.2 Order

Definition 1.7 (Order of a group). For a finite group G , we call $|G|$ the number of elements in G , its order.

Remark 1.8 (Groups of Order 2 and 3). These are the unique groups of order 2 and 3.

	e	a		e	a	b
	e	a		e	a	b
e	e	a		a	b	e
a	a	e		b	e	a

$$G_2 = \{e, a\}$$

$$G_3 = \{e, a, b\}$$

There are no trivial subgroups.

Example 1.9 (Groups of Order 4).

$$V_4 = \{e, a, b, c\}$$

$\cdot$	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

What are its subgroups?

We will see: If G is finite and $H \leq G$, then $|H|$ divides $|G|$.

Theorem 1.10 (Checking for Subgroups) *A subset H of G is a subgroup if*

- H is closed under the operation of G ;
- the identity element e of G is in H ;
- for any $a \in H$, the inverse a^{-1} is also in H .

Proof. H needs an identity to be a subgroup, and that identity element must be e , the same as G because otherwise

$$e_H * h = h$$

would not imply that the identity of H is the identity of G .

Associativity works here because the induced operation is associative.

The inverse of $a \in H$ is a^{-1} . Just prove this by assuming that a has an alternate inverse a_H . $\square$

Definition 1.11 (General Linear Group). The General Linear Group $GL_n(\mathbb{C})$ is the set of $n \times n$ invertible matrices with entries in $\mathbb{C}$.

1.3 Linear Subgroups

Definition 1.12 (Simple Linear Group). Define the Special Linear Group $SL_n(\mathbb{C})$ as the set of $n \times n$ invertible matrices with entries in $\mathbb{C}$ with determinant 1.

Proposition 1.13 (subgroup) *We claim the Simple Linear Group is a subgroup of GL_n* Proposition

Proof. We see immediately that $SL_n(\mathbb{C}) \subseteq GL_n(\mathbb{C})$, since $\det \neq 0 \implies$ invertible

Then we see that if $\det(A) = \det(B) = 1 \implies \det(AB) = \det(A)\det(B) = 1$, establishing closure

We see the identity I where we see $\det I = 1$

Lastly, we look at the inverse, where we see $\det A^{-1} = \frac{1}{\det A} = 1$, so it's closed under inverses $\square$

1.4 Cyclic Subgroups

We have for $a \in G$, let $a^n = a * a * \dots * a$ n times. Where $a^{-1} = a^1$, $a^{-n} = (a^{-1})^n$. Let $a^0 = e$.

Theorem 1.14 (Cyclic Subgroups) *Let a be an element of a group G . Then*

$$H = \{a^n | n \in \mathbb{Z}\}$$

is a subgroup of G , and every subgroup of G which contains a must contain H .

Proof. We see that $a^n * a^m = a^{n+m}$ pretty easily from associativity.

Then $e = a^0 \in H$.

Lastly $(a^n)^{-1} = a^{-n} \in H$ $\square$

Definition 1.15 (Group "Generation"). An element $a \in G$ generates G is $\langle a \rangle = G$.

G is cyclic if there is some element a that generates it.

2 Definitions and notation

Definition 2.1 (First definition). Write the definition and one sentence explaining why it matters.

3 Claims, theorems, and proof sketches

Claim. State the claim in one precise sentence.

Proof sketch. Record the key idea, the first useful equation, and the step that still needs checking.

4 Examples and calculations

Example 4.1 (Lecture example). Write the setup, the calculation, and the conclusion.

5 Questions and follow-up

- What is still unclear?
- Which exercise or reading should I revisit?
- TODO: